



Oran Duan

Beijing, China

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EDUCATION

Communication University of China | Beijing

Master of Science in Intelligent Media and Immersive Technology | Sep 2023 - Jul 2026 (Expected)

- **Thesis:** Multimodal Autoregressive Dance Motion Generation Model (3D Motion Generation, Computer Vision, Diffusion Model, AIGC)

Bachelor of Engineering in Digital Media Technology (Intelligent Media Track) | Sep 2019 - Jul 2023

- **Thesis:** Augmented Reality System Implementation based on ORB-SLAM and Unity (SLAM Algorithm, AR Application, Unity)

PROJECTS

Audio-Driven Lip Sync for Digital Humans | National Invention Patent (First Inventor) | Dec 2024 - Oct 2025

- **Challenge:** Addressed the need for realistic and accurate audio-to-lip-sync in digital avatars by solving multimodal alignment issues.
- **Action:** Designed a novel contrastive loss function to minimize the similarity between audio features and expression coefficients, achieving precise modality alignment. Led the entire model development lifecycle, from architecture and training to optimization.
- **Results:** Successfully filed a **national invention patent** as the **first inventor**, establishing a **core technical moat**. Achieved a 50% reduction in model parameters and a 20% increase in inference speed compared to leading open-source models, enabling low-latency, real-time applications.

Peking Opera Facial Makeup Generation | SCI Q1 Journal Paper (2nd Author & Corresponding Author) | Sep 2023 - Oct 2024

- **Challenge:** Developed a high-fidelity system to generate stylized and characteristic Peking Opera facial makeup, overcoming issues of poor style control and low accuracy in existing methods.
- **Action:** Based on StyleGAN2, architected a co-training network and a style-conditioned transformation network. Led data augmentation, dataset collection, and hyperparameter tuning.
- **Results:** Achieved state-of-the-art performance with an **FID score of 13.43** and **KID $\times 10^3$ of 6.47**, an improvement of **38% and 57%** respectively over baseline models. Published findings in **Heritage Science (CAS Q1, Arts & Humanities)**, providing a validated technical solution for digital cultural heritage preservation.

Fitness Dance Scoring System | National Innovation Project (Project Leader) | Mar 2022 - Oct 2023

- **Challenge:** Solved the lack of objective, quantitative scoring tools in fitness dance education by developing a pose recognition-based system.
- **Action:** As Project Leader, designed a dance evaluation algorithm using OpenPose and built a full user interaction system. Optimized performance on a consumer-grade GTX 1060 GPU to balance accuracy and real-time processing.
- **Results:** Achieved real-time processing at **8.2-9.5 FPS with "Good" recognition** accuracy on consumer-grade hardware. The DTW-based scoring system was validated with a **peak score of 85/100**. Secured a **Third Prize** at the **8th "Internet+" Innovation Competition (Beijing Division)** and published an **EI-indexed conference paper**.

INTERNSHIP

Zhipu AI | Beijing

Digital Human Technology Algorithm Intern | Nov 2024 - Sep 2025

- Managed the algorithm team's Linux server environments and GitLab repositories.
- Assisted in training and iterating on audio-driven video lip-sync models, contributing to an internal patent and technical documentation.
- Researched and replicated face-swapping algorithms to support the launch of a ToB e-commerce digital human live-streaming application.
- Collaborated with CogVideo and CodeGeex teams on data and algorithm integration.

Migu Video (Paris Olympics 2024) | Beijing

Broadcast Data Packaging Intern | Aug 2024 - Sep 2024

- Packaged and stylized live sports data for broadcast during the Paris Olympics, ensuring seamless integration with commercial AR animations and subtitles.
- Acted as the liaison between Migu Video and Beijing Television, guaranteeing data accuracy and broadcast quality.

Beijing 2022 Winter Olympics | Beijing & Zhangjiakou

BTP Broadcast Technology Trainee (Liaison Officer) | Feb 2022 - Apr 2022

- Selected for the OBS Broadcast Training Program and served as a Liaison Officer at the Big Air Shougang and Zhangjiakou Paralympic Village.
- Supported live broadcast system management, assisted rights-holding broadcasters, and managed the mixed zone operations.

CAMPUS ACTIVITY

Communication University of China | Beijing

Student Assistant & Deputy Editor, Media Center | Oct 2022 - Jun 2024

- Managed student affairs, including award applications and archival work for the Student Affairs Office.
- As Deputy Editor, oversaw content review, public account operations, and team development for the university's official media.

Class President & Youth League Secretary, Class of 2019 | Sep 2020 - Jun 2023

- Led the class to win multiple honors, including "Outstanding Class Collective" and the "Beijing May Fourth Red Flag Youth League Branch."

HONORS& AWARDS

- **National Scholarship (Graduate Level)** | 2024-2025
- **Beijing Municipal Outstanding Graduate** | 2023
- **Outstanding Student Leader (Undergraduate)** | 2023
- **National Encouragement Scholarship** | 2019-2023 (3 years)

SKILLS & INTERESTS

- **Languages:** Python, C/C++, MATLAB
- **Frameworks:** PyTorch, CUDA
- **Tools & Software:** Unity, ComfyUI, Premiere, Audition, Notion, Office Suite
- **Human Languages:** Mandarin (Native), English (CET-6, Proficient in technical literature)
- **Other Interests:** Badminton, Street Dance, PC game.